



WIP: WORKSHOP ON INTERPRETABILITY IN PROGRESS

UNDERSTANDING, FORECASTING, AND REPAIRING TRAINING

Isabelle Lee
USC

Emmy Liu
CMU

Kaiser Sun
JHU

Daniel Khoshdel
JHU

Stella Biderman
Eleuther

David Alvarez-Melis
Harvard

ABSTRACT

Interpretability research has produced a robust toolkit for understanding models, but these tools are often only applied to models at the end of training. The path that a model takes through training shapes what it becomes, and a trajectory-level understanding may enable us to trace how its behaviors emerge, anticipate failures and safety risks, and intervene before they fully manifest. This workshop asks which interpretability methods work for understanding training, and what actions might they enable. We organize our program around three escalating goals. (1) *Descriptive* work describes how representations and model internals evolve over training. (2) *Predictive* work turns this understanding proactive, identifying early signals that forecasts later capabilities, failures, generalization, and other structural shifts in the model. (3) *Prescriptive* work uses those insights to control and repair training via checkpoint-level interventions with data mixing and filtering strategies, or by directly steering learning with methods such as gradient routing and targeted regularization strategies. We welcome both theoretical and empirical work that aligns with these goals. By bringing together academic researchers and practitioners, the workshop aims to move from understanding training to foresight and intervention.

Tagline. Toward actionable training dynamics: describe trajectories, predict outcomes, repair training.

Website. <https://science-of-unfinished-models.github.io/>

1 WORKSHOP DESCRIPTION

Interpretability research has produced a wide-ranging toolkit, with probing, sparse dictionary learning [1; 2], and data attribution [3; 4; 5; 6], but these methods are mostly applied to finished models. How a model gets there is additionally studied through training dynamics and with other prediction methods like scaling laws. Aggregate statistical relationships such as scaling laws [7; 8; 9; 10; 11] and architecture-based performance estimation [12; 13; 14] reliably predict final losses, while studies of training dynamics can document *when* representations and behaviors emerge over training steps [15], across scale [16]. These lines of work establish that the path a model takes through training shapes what it becomes. Then, a relevant question, and the focus of this workshop, becomes *which interpretability methods work for understanding training, and what actions can they enable that outcome-only evaluation cannot?*

A trajectory-level view of training is useful precisely because it surfaces phenomena that final perplexity, accuracy, or win-rates cannot. For example, models may perform well on downstream evaluations while becoming overtrained, making them difficult to fine-tune subsequently [17; 18]. If model flexibility could be quantified *during* training, such failure modes could be anticipated before they manifest downstream. In a similar vein, recent work on the interaction between fine-tuning and pretraining suggests that training dynamics can predict when adaptation will succeed, degrade, or induce forgetting [19; 20; 21]. Bridging the gap between understanding and practice requires moving beyond conceptual descriptions toward diagnosing failures that outcome-only evaluation misses and toward guiding interventions during training.

This workshop aims to encourage such types of work that can provide insights toward a dynamical and actionable understanding of training [22; 23; 24]. We organize this around three escalating goals:

1. **Descriptive:** characterizing how representations, circuits, gradients, and behaviors evolve over the course of training.
2. **Predictive:** identifying signals early in training that reliably forecast later capabilities, failures, generalization, or structural changes.
3. **Prescriptive:** using predictive insight to actively steer, control, and repair training.

We organize the workshop’s topics of interest around the three goals detailed below and welcome work that shares similar interests:

1. Descriptive

- Representation formation, phase transitions, reorganization, and interaction across training [25; 26; 15; 27; 28]
- Interpretability across time, scale, gradients, and optimization dynamics [29; 30; 31]
- Characterizing the impact of initializations and run-to-run variations [32; 33; 34]

2. Predictive

- Pretraining evaluations that predict post-training behavior [35; 36; 37]
- Generalization prediction: what transfers from which data, when, and why [38; 39; 40]
- Training outcome prediction, including negative results [41; 42; 43; 10]

3. Prescriptive

- Training repair, such as reverting to and correcting from earlier checkpoints [44; 45; 46; 47; 48]
- Phase specific data selection strategies [49; 44; 50]
- Controlling model behavior during training [51; 52; 53]

Relevant Workshops Our proposed workshop is closely related to the High-dimensional Learning Dynamics (HiLD) workshop series [54; 55; 56; 57], overlapping with our *descriptive* layer: representation formation, phase transitions, and interpretability across training. However, we substantially broaden this scope to include predicting from early failure signals and possible training repairs—what could make the science of training actionable [23]. Beyond descriptive analysis, we emphasize *predictive* and *prescriptive* dimensions of training dynamics, seeking work in both theory as well as how the theoretical insights inform practice, beyond HiLD’s theoretical focus. We highlight the need for methods that *anticipate* training dynamics and outcomes (e.g., early detection of failure modes, generalization prediction, and checkpoint stress-testing) and that enable actions such as training repair, causal manipulation, and phase-aware data selection. A central goal is to bring together academic researchers and practitioners to identify which analyses are useful in practice, where current methods fall short, and what predictive or prescriptive tools would best support real-world model development.

2 INVITED SPEAKERS AND PANELISTS

Below are the currently confirmed speakers and panelists. We intend to invite an additional PhD student speaker.

Andrew Saxe <a.saxe@ucl.ac.uk> is a professor at UCL. His lab develops mathematical and experimental tools to understand how learning emerges in brains and artificial neural networks.

Gautam Reddy <greddy@princeton.edu> is an assistant professor at Princeton, whose lab develops physics-inspired theory and tools for understanding learning and decision-making in biological and artificial systems.

Kyle Lo <kyleclo@cs.washington.edu> is a member of technical staff at Microsoft AI. Previously, he co-lead the OLMo project on open language modeling. His current research focuses on large-scale pretraining of language models, with an emphasis on data curation, data mixing, benchmark validity, efficient experimentation, and generalization across model scales.

Martin Ziqiao Ma <marstin@umich.edu> is a Member of Technical Staff at Thinking Machines Lab. His research focuses on continual learning with minimal and natural supervision.

Stephanie Chan <chan.stephanie.cy@gmail.com> is a staff research scientist at Google Deepmind. Her current research focuses on the future impacts of AI and science of AI.

3 TENTATIVE SCHEDULE AND LOGISTICS

Tentative important dates for submissions, reviews, and final materials: We invite short (5-page) or long (9-page) paper submissions. Submissions will be managed via OpenReview.

- Paper submission deadline: February 1, 2027
- Reviewer assignment period: February 2 – February 4, 2027
- Review deadline: February 18, 2027
- Acceptance/rejection notification: February 24, 2027
- Accepted papers public on OpenReview and imported to iclr.cc: February 26, 2027
- Camera-ready deadline: March 19, 2027
- Workshop schedule (opening/closing times) posted on iclr.cc: March 26, 2027
- Final workshop program finalized in organizer portal: April 14, 2027
- Workshop date: April 29 or 30, 2027 (TBD by ICLR)

Tentative schedule. We propose a full-day workshop consisting of invited talks, contributed oral presentations selected from accepted submissions, two poster sessions, and a combined panel and interactive discussion designed to foster community engagement.

Time	Event	Time	Event
9:00–9:10	Opening Remarks	13:30–14:10	Invited Talk 4
9:10–9:50	Invited Talk 1	14:10–14:50	Contributed Orals
9:50–10:30	Invited Talk 2	14:50–15:30	Poster Session 2
10:30–11:15	Poster Session 1	15:30–16:00	Panel Discussion
11:15–11:55	Invited Talk 3	16:00–17:00	Interactive “Debugging Training” Session
11:55–12:45	Contributed Orals	17:00–17:10	Closing Remarks
12:45–13:30	Lunch		

The closing hour combines a moderated panel discussion on open challenges in the field with an interactive “debugging training runs” segment, where participants bring real training bugs—instabilities, divergences, scaling pathologies—for the panelists and audience to diagnose together. We will collect training bug examples through a short survey shortly before the workshop, then split into small-group discussions led by organizers and panelists. In the final 20 minutes, each group will share key takeaways and highlights from their discussion. Following the workshop, we will host an informal social for organizers, speakers, and participants.

3.1 PARTICIPATION AND ACCESSIBILITY PLAN

At ICLR 2027, we anticipate receiving approximately 100 paper submissions, with an expected acceptance rate of around 35–45%. Of the accepted papers, approximately 20–30% will be invited for contributed oral presentations, with the remainder presented as posters. As this is a first-time workshop, prior submission, acceptance, and attendance statistics are not available.

Attendees. We expect 100–200 in-person attendees. Our goal is to attract a diverse range of participants, including: 1) graduate students and early-career researchers, through accessible workshop materials, mentorship-oriented programming, and opportunities for contributed presentations; 2) researchers from underrepresented and interdisciplinary communities, through broad outreach, inclusive call-for-papers language, and targeted dissemination through relevant Slack and Discord communities, EleutherAI channels, social media platforms such as X/Twitter and Bluesky; and 3) senior researchers and practitioners, through invited talks, panel discussions, and opportunities for collaboration.

Virtual Access and Outreach. To broaden participation, we will maintain an online presence through the workshop website, social channels, academic mailing lists, and relevant research communities. Where permitted, accepted papers, posters, and presentation materials will be posted online, along with summaries and key discussions, to support participants unable to attend physically and engage the broader community.

Diversity of Organizers and Speakers. Our organizing team reflects this commitment, spanning diverse career stages, disciplines, institution types, geographies, and lived experiences, including nontra-

ditional academic pathways and underrepresented gender identities. Our proposed speakers and panelists similarly represent a range of seniority levels, research communities, and institutional contexts. As we finalize invitations, we will continue prioritizing diversity across lived experiences and career.

Special Requirements & Technical Needs. No special needs beyond the standard ICLR workshop setup. We request poster space for two poster sessions and standard AV support, including projector/display, microphones, reliable internet, live streaming, and recording where supported by ICLR.

4 ORGANIZERS AND PROGRAM COMMITTEE

Below are the organizers bio and a list of confirmed program committee. To the best of our knowledge, none of the organizers are listed on any other ICLR 2027 workshop proposals. Organizers will not assess submissions from their own institutions or from individuals with whom they hold personal conflicts of interest, and any such submissions will be handled solely by non-conflicted organizers and reviewers.

Isabelle Lee <lee.isabelle.g@gmail.com> is a PhD student at the University of Southern California. Her work focuses on interpretability, training, and reasoning. Specifically, her research aims to (1) predict training and (2) predict failures. To this aim, she uses dynamical systems and physics-inspired approaches to analyze small-scale toy learning problems, as well as analyzing larger-scale training from developmental perspectives. She is a recipient of the Viterbi School of Engineering Graduate Fellowship and Coefficient Giving’s Technical AI Safety Research Grant. Previously, she was a researcher at Carnegie Mellon University and Amazon Alexa, and her background is in Plasma Physics and Complex Systems.

Emmy Liu <emmy@cmu.edu> is a PhD Candidate at Carnegie Mellon University in the Language Technologies Institute. Her current work is on understanding pretraining from a cognitive perspective. She is a recipient of a Natural Sciences and Engineering Research Council of Canada Doctoral Scholarship, the SoftBank-ARM fellowship, an NEC fellowship as well as a CMU Presidential Fellowship. Her work has received the Best Resource Paper award at ACL 2023. She has co-organized the workshop on Figurative Language Processing at NAACL 2024, the overall Student Research Workshop at NAACL 2025, as well as the Computational Developmental Linguistics workshop at ACL 2026.

Kaiser Sun <hsun74@jhu.edu> is a Ph.D. student in Computer Science at the Data Science and AI Institute and the Center for Language and Speech Processing at Johns Hopkins University. Their research focuses on the training dynamics, evaluation, and interpretability of Large Language Models. Generally, Kaiser is interested in understanding how language models function and exploring how we can change them. Their work has been recognized with an honorable mention at CoNLL2023, an outstanding paper award at MLRC2022, and coverage by a few media outlets such as MIT Technology Review and WIRED. Previously, they obtained their B.S. and M.S. from the University of Washington and were a researcher at Meta AI, Together AI, and AWS AI Labs.

Daniel Khashabi <danielk@jhu.edu> is an Assistant Professor of Computer Science at Johns Hopkins University, and is affiliated with the Data Science and AI Institute and the Center for Language and Speech Processing (CLSP). He is interested in the intersection of reasoning and discovery. He is the recipient of the NSF CAREER award (2026), Ai2’s Lasting Impact Award (2024), multiple industry faculty awards (Amazon, Google, Apple), and several Best Paper Awards (COLM 2024, ACL 2023, NAACL 2022). His teaching has earned him the Joel Dean Award for Excellence in Teaching in the Department of Computer Science in 2025. Most relevant to this proposal, he has served as a co-organizer of Towards Test-Time Continual Learning Agents at NeurIPS 2026, Interpreting Agent Behavior (IAB) at NeurIPS 2026, the Workshop on Instruction Tuning and Instruction Following at NeurIPS 2023, and the Student Research Workshop at ACL 2019.

Stella Biderman <stella@eleuther.ai> is the Executive Director of EleutherAI, a non-profit research institute and open science community that does research on large language models, works to make research on large-scale AI technologies more widely accessible around the world, and works to grow and promote the open source AI community. She pioneered the development of open and transparent language models that make the type of research advocated for in this workshop possible,

including the first state-of-the-open LLMs to be released alongside their training data [58; 59; 60] and the Pythia model suite [47] which introduced the idea that releasing partially trained model checkpoints was scientifically valuable and set the standard for maximally open releases adopted by projects such as LLM360, OLMo and Marin. Her lab has done influential research on learning dynamics and interpretability, including introducing Sparse Autoencoders as a tool for interpretability [61], introducing the problem of forecasting which specific sequences are memorized by a large language model during training [62], performing the first study of the stability of circuit analysis over the course of training [63], and demonstrating that it is possible to use data filtering to eliminate undesirable capabilities in large language models [49]. Her recent position paper at ICML outlines the need for the type of work solicited in this workshop [22].

Stella co-leads the EvalEval Coalition, a multi-stakeholder initiative bringing methodological rigor and accountability to AI evaluation, is on the advisory board to the Machine Learning Reproducibility Challenge, is organizing an Open-Weight Pre-training Safety Accelerator as part of the Seoul Alignment Workshop (co-located with ICML), and is an AC for NeurIPS. Previously, she was an organizer of the AI Village @ DEF CON (2018-2021), working group lead for the BigScience Research Workshop (2022), and organized Mozilla and EleutherAI’s Dataset Convening. Stella received her M.S. in Computer Science from Georgia Tech, her S.B. (honors) in Mathematics from the University of Chicago, and her A.B. in Philosophy from the University of Chicago. Her work regularly appears at NeurIPS, ICML, ICLR, and ACL.

David Alvarez-Melis <dam@seas.harvard.edu> is an Assistant Professor of Computer Science at Harvard University, where he leads the Data-Centric Machine Learning (DCML) Lab, with appointments as Associate Faculty at the Kempner Institute and Research Scientist at Microsoft Research. His group develops principled methods for characterizing, curating, and optimizing training data, work that bears directly on the workshop’s prescriptive axis—in particular, treating data selection, ordering, and composition as levers for shaping training trajectories and downstream behavior. He has organized NeurIPS-scale workshops at the intersection of machine learning and applied mathematics, most directly comparable being the Optimal Transport and Machine Learning Workshop at NeurIPS 2023, along with the Optimal Transport Workshop at the Institut d’Études Scientifiques de Cargèse (2024) and the LatinX in AI Workshop at ICML (2023, 2024). David received his PhD in Computer Science from MIT, his MS in Mathematics from NYU’s Courant Institute, and his BSc in Mathematics from ITAM in Mexico City. His work appears regularly at NeurIPS, ICML, ICLR, AISTATS, and UAI.

4.1 PROGRAM COMMITTEE

We are continuing to recruit program committee members with expertise across the workshop themes and will aim for diversity across institutions, geographies, career stages, and research areas. We anticipate approximately 100 submissions. Each paper will receive 3 reviews, with no reviewer assigned more than 3 papers, requiring a program committee of approximately 100 members. **To manage conflicts of interest**, submissions will be assigned so that no reviewer or organizer evaluates a paper from their own institution or from anyone with whom they have a personal conflict (e.g., advisor–advisee, close collaborators, or family), and organizers with a conflict on a given submission will be recused from its decision. Our current program committee (22 confirmed) includes:

Sarah Liaw (Harvard), Millicent Li (Northeastern), Leonardo Blas Urrutia (USC), Terry Zhou (Harvard), Indranil Halder (Harvard), Jennifer Lu (Brown/MIT), Ruochen Zhang (Brown/Microsoft), Rachit Bansal (Harvard), Logan Riggs-Smith (AISST), Alice Rigg (NDIF), Jeffrey Andrade (Harvard), Jay Huang (JHU), Bernal Jimenez Gutierrez (JHU), Niyati Bafna (JHU), Bingbin Liu (Harvard), Marc Marone (JHU), Elisabeth Fittschen (JHU), Leo Du (JHU), Glenn Matlin (Georgia Tech), Xiaochuan Li (CMU), Atharva Naik (CMU), Dabashish Chakraborty (JHU)

We also plan to host an after-social for organizers, program committee, participants and speakers, organized by our social chair, Glenn Matlin (Georgia Tech).

4.2 ADVISORY COMMITTEE

Confirmed: Zhuoran Lu (Hong Kong Univeristy), Jen-Tse Huang (JHU), Jie Gao (Monash University)

REFERENCES

- [1] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- [2] Usha Bhalla, Alex Oesterling, Claudio Mayrink Verdun, Himabindu Lakkaraju, and Flavio P. Calmon. Temporal sparse autoencoders: Leveraging the sequential nature of language for interpretability, 2026. URL <https://arxiv.org/abs/2511.05541>.
- [3] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions, 2020. URL <https://arxiv.org/abs/1703.04730>.
- [4] Jiacheng Liu, Taylor Blanton, Yanai Elazar, Sewon Min, YenSung Chen, Arnavi Chheda-Kothary, Huy Tran, Byron Bischoff, Eric Marsh, Michael Schmitz, Cassidy Trier, Aaron Sarnat, Jenna James, Jon Borchardt, Bailey Kuehl, Evie Cheng, Karen Farley, Sruthi Sreeram, Taira Anderson, David Albright, Carissa Schoenick, Luca Soldaini, Dirk Groeneveld, Rock Yuren Pang, Pang Wei Koh, Noah A. Smith, Sophie Lebrecht, Yejin Choi, Hannaneh Hajishirzi, Ali Farhadi, and Jesse Dodge. Olmotrace: Tracing language model outputs back to trillions of training tokens, 2025. URL <https://arxiv.org/abs/2504.07096>.
- [5] Glenn Matlin, Chandreyi Chakraborty, Saehee Eom, Mika Okamoto, Rayan Castilla, Louis Jaburi, Alvin Deng, Taywon Min, Lucia Quirke, Stella Biderman, and Mark Riedl. Capability provenance in language models: A case study in social reasoning, 2026. URL <https://arxiv.org/abs/2606.19625>.
- [6] Sunwoo Kim, Seokwon Jung, Sohyung Kim, Seong Joon Oh, and Alice Oh. Form over content in gradient-based data attribution methods, 2026. URL <https://arxiv.org/abs/2609.19589>.
- [7] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL <https://arxiv.org/abs/2001.08361>.
- [8] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL <https://arxiv.org/abs/2203.15556>.
- [9] Yasaman Bahri, Ethan Dyer, Jared Kaplan, Jaehoon Lee, and Utkarsh Sharma. Explaining neural scaling laws. *Proceedings of the National Academy of Sciences*, 121(27), June 2024. ISSN 1091-6490. doi: 10.1073/pnas.2311878121. URL <http://dx.doi.org/10.1073/pnas.2311878121>.
- [10] Ethan Caballero, Kshitij Gupta, Irina Rish, and David Krueger. Broken neural scaling laws, 2023. URL <https://arxiv.org/abs/2210.14891>.
- [11] Tim Pearce and Jinyeop Song. Reconciling kaplan and chinchilla scaling laws, 2024. URL <https://arxiv.org/abs/2406.12907>.
- [12] Colin White, Arber Zela, Binxin Ru, Yang Liu, and Frank Hutter. How powerful are performance predictors in neural architecture search?, 2021. URL <https://arxiv.org/abs/2104.01177>.
- [13] Thomas Elsken, Jan Hendrik Metzen, and Frank Hutter. Neural architecture search: a survey. *J. Mach. Learn. Res.*, 20(1):1997–2017, January 2019. ISSN 1532-4435.

- [14] Joseph Mellor, Jack Turner, Amos Storkey, and Elliot J. Crowley. Neural architecture search without training, 2021. URL <https://arxiv.org/abs/2006.04647>.
- [15] Angelica Chen, Ravid Shwartz-Ziv, Kyunghyun Cho, Matthew L. Leavitt, and Naomi Saphra. Sudden drops in the loss: Syntax acquisition, phase transitions, and simplicity bias in mlms, 2025. URL <https://arxiv.org/abs/2309.07311>.
- [16] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models, 2022. URL <https://arxiv.org/abs/2206.07682>.
- [17] Jacob Mitchell Springer, Sachin Goyal, Kaiyue Wen, Tanishq Kumar, Xiang Yue, Sadhika Malladi, Graham Neubig, and Aditi Raghunathan. Overtrained language models are harder to fine-tune, 2025. URL <https://arxiv.org/abs/2503.19206>.
- [18] Shibhansh Dohare, J Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A Rupam Mahmood, and Richard S Sutton. Loss of plasticity in deep continual learning. *Nature*, 632 (8026):768–774, 2024.
- [19] Yi Ren and Danica J. Sutherland. Learning dynamics of llm finetuning, 2025. URL <https://arxiv.org/abs/2407.10490>.
- [20] Kaiser Sun and Mark Dredze. Amuro & char: Analyzing the relationship between pre-training and fine-tuning of large language models. In Vaibhav Adlakha, Alexandra Chronopoulou, Xiang Lorraine Li, Bodhisattwa Prasad Majumder, Freda Shi, and Giorgos Vernikos, editors, *Proceedings of the 10th Workshop on Representation Learning for NLP (Repl4NLP-2025)*, pages 131–151, Albuquerque, NM, May 2025. Association for Computational Linguistics. ISBN 979-8-89176-245-9. doi: 10.18653/v1/2025.repl4nlp-1.11. URL <https://aclanthology.org/2025.repl4nlp-1.11/>.
- [21] Louis Bethune, David Grangier, Dan Busbridge, Eleonora Gualdoni, Marco Cuturi, and Pierre Ablin. Scaling laws for forgetting during finetuning with pretraining data injection, 2025. URL <https://arxiv.org/abs/2502.06042>.
- [22] Stella Biderman, Mohammad Aflah Khan, Niloofar Miresghallah, Catherine Arnett, Fazl Barez, and Naomi Saphra. Position: Don’t just “fix it in post”: A science of ai must study training dynamics. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 205 of *Proceedings of Machine Learning Research*, 2026.
- [23] Hadas Orgad, Fazl Barez, Tal Haklay, Isabelle Lee, Marius Mosbach, Anja Reusch, Naomi Saphra, Byron Wallace, Sarah Wiegrefe, Eric Wong, et al. Interpretability can be actionable. *arXiv preprint arXiv:2605.11161*, 2026.
- [24] Isabelle Lee, Emmy Liu, Cathy Jiao, Brihi Joshi, Dani Yogatama, Fazl Barez, and Michael Saxon. Rigorous interpretation is a form of evaluation, 2026. URL <https://arxiv.org/abs/2605.05508>.
- [25] Yang Xu, Yi Wang, Hengguan Huang, and Hao Wang. Tracking the feature dynamics in llm training: A mechanistic study, 2025. URL <https://arxiv.org/abs/2412.17626>.
- [26] Deniz Bayazit, Aaron Mueller, and Antoine Bosselut. Crosscoding through time: Tracking emergence & consolidation of linguistic representations throughout llm pretraining, 2026. URL <https://arxiv.org/abs/2509.05291>.
- [27] Alessandro Achille, Matteo Rovere, and Stefano Soatto. Critical learning periods in deep neural networks, 2019. URL <https://arxiv.org/abs/1711.08856>.
- [28] Leo Z. Liu, Yizhong Wang, Jungo Kasai, Hannaneh Hajishirzi, and Noah A. Smith. Probing across time: What does roberta know and when?, 2021. URL <https://arxiv.org/abs/2104.07885>.

- [29] Jianhui Chen, Yuzhang Luo, and Liangming Pan. Mechanistic data attribution: Tracing the training origins of interpretable llm units, 2026. URL <https://arxiv.org/abs/2601.21996>.
- [30] Tyler A. Chang, Zhuowen Tu, and Benjamin K. Bergen. Characterizing learning curves during language model pre-training: Learning, forgetting, and stability, 2024. URL <https://arxiv.org/abs/2308.15419>.
- [31] Henry C. Conklin, Tom Hosking, Tan Yi-Chern, Julian Gold, Jonathan D. Cohen, Thomas L. Griffiths, Max Bartolo, and Seraphina Goldfarb-Tarrant. Learning is forgetting: Llm training as lossy compression, 2026. URL <https://arxiv.org/abs/2604.07569>.
- [32] Clara Meister, Wojciech Stokowiec, Tiago Pimentel, Lei Yu, Laura Rimell, and Adhiguna Kuncoro. A natural bias for language generation models, 2023. URL <https://arxiv.org/abs/2212.09686>.
- [33] Alexander D’Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, Farhad Hormozdiari, Neil Houlsby, Shaobo Hou, Ghassen Jerfel, Alan Karthikesalingam, Mario Lucic, Yian Ma, Cory McLean, Diana Mincu, Akinori Mitani, Andrea Montanari, Zachary Nado, Vivek Natarajan, Christopher Nielson, Thomas F. Osborne, Rajiv Raman, Kim Ramasamy, Rory Sayres, Jessica Schrouff, Martin Seneviratne, Shannon Sequeira, Harini Suresh, Victor Veitch, Max Vladymyrov, Xuezhi Wang, Kellie Webster, Steve Yadlowsky, Taedong Yun, Xiaohua Zhai, and D. Sculley. Underspecification presents challenges for credibility in modern machine learning, 2020. URL <https://arxiv.org/abs/2011.03395>.
- [34] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping, 2020. URL <https://arxiv.org/abs/2002.06305>.
- [35] Arkil Patel, Siva Reddy, Marius Mosbach, and Dzmitry Bahdanau. Forecasting downstream performance of llms with proxy metrics, 2026. URL <https://arxiv.org/abs/2605.18607>.
- [36] Fan Chen, Audrey Huang, Noah Golowich, Sathika Malladi, Adam Block, Jordan T. Ash, Akshay Krishnamurthy, and Dylan J. Foster. The coverage principle: How pre-training enables post-training, 2025. URL <https://arxiv.org/abs/2510.15020>.
- [37] Jiaxin Wen, Zhengxuan Wu, Dawn Song, and Lijie Chen. Generalization dynamics of lm pre-training, 2026. URL <https://arxiv.org/abs/2609.33150>.
- [38] Emmy Liu, Kaiser Sun, Millicent Li, Isabelle Lee, Lindia Tjauatja, Jen tse Huang, and Graham Neubig. What do language models learn and when? the implicit curriculum hypothesis, 2026. URL <https://arxiv.org/abs/2604.08510>.
- [39] Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. Doremi: Optimizing data mixtures speeds up language model pretraining, 2023. URL <https://arxiv.org/abs/2305.10429>.
- [40] Jaydeep Borkar, Karan Chadha, Niloofar Miresghallah, Yuchen Zhang, Irina-Elena Veliche, Archi Mitra, David A. Smith, Zheng Xu, and Diego Garcia-Olano. Memorization dynamics in knowledge distillation for language models, 2026. URL <https://arxiv.org/abs/2601.15394>.
- [41] Nicholas Lourie, Michael Y. Hu, and Kyunghyun Cho. Scaling laws are unreliable for downstream tasks: A reality check, 2025. URL <https://arxiv.org/abs/2507.00885>.
- [42] Yangyi Chen, Binxuan Huang, Yifan Gao, Zhengyang Wang, Jingfeng Yang, and Heng Ji. Scaling laws for predicting downstream performance in llms, 2025. URL <https://arxiv.org/abs/2410.08527>.

- [43] Deep Ganguli, Danny Hernandez, Liane Lovitt, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova Dassarma, Dawn Drain, Nelson Elhage, Sheer El Showk, Stanislav Fort, Zac Hatfield-Dodds, Tom Henighan, Scott Johnston, Andy Jones, Nicholas Joseph, Jackson Kernian, Shauna Kravec, Ben Mann, Neel Nanda, Kamal Ndousse, Catherine Olsson, Daniela Amodei, Tom Brown, Jared Kaplan, Sam McCandlish, Christopher Olah, Dario Amodei, and Jack Clark. Predictability and surprise in large generative models. In *2022 ACM Conference on Fairness Accountability and Transparency, FAccT '22*, page 1747–1764. ACM, 2022. doi: 10.1145/3531146.3533229. URL <http://dx.doi.org/10.1145/3531146.3533229>.
- [44] Cameron Tice, Puria Radmard, Samuel Ratnam, Andy Kim, David Africa, and Kyle O’Brien. Alignment pretraining: Ai discourse causes self-fulfilling (mis)alignment, 2026. URL <https://arxiv.org/abs/2601.10160>.
- [45] Alex Cloud, Jacob Goldman-Wetzler, Evžen Wybitul, Joseph Miller, and Alexander Matt Turner. Gradient routing: Masking gradients to localize computation in neural networks, 2024. URL <https://arxiv.org/abs/2410.04332>.
- [46] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning, 2020. URL <https://arxiv.org/abs/2001.06782>.
- [47] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023. URL <https://arxiv.org/abs/2304.01373>.
- [48] Neil Rathi and Alec Radford. Shaping capabilities with token-level data filtering, 2026. URL <https://arxiv.org/abs/2601.21571>.
- [49] Kyle O’Brien, Stephen Casper, Quentin Anthony, Tomek Korbak, Robert Kirk, Xander Davies, Ishan Mishra, Geoffrey Irving, Yarin Gal, and Stella Biderman. Deep ignorance: Filtering pretraining data builds tamper-resistant safeguards into open-weight llms, 2025. URL <https://arxiv.org/abs/2508.06601>.
- [50] Ian Magnusson, Nguyen Tai, Ben Bogin, David Heineman, Jena D. Hwang, Luca Soldaini, Akshita Bhagia, Jiacheng Liu, Dirk Groeneveld, Oyvind Tafjord, Noah A. Smith, Pang Wei Koh, and Jesse Dodge. Datadecide: How to predict best pretraining data with small experiments, 2025. URL <https://arxiv.org/abs/2504.11393>.
- [51] Viktor Moskvoretskii, Dominik Glandorf, Jorge Medina Moreira, Tanja Käser, and Robert West. Tracing persona vectors through llm pretraining, 2026. URL <https://arxiv.org/abs/2605.13329>.
- [52] Helena Casademunt, Caden Juang, Adam Karvonen, Samuel Marks, Senthoooran Rajamanoharan, and Neel Nanda. Steering out-of-distribution generalization with concept ablation fine-tuning, 2025. URL <https://arxiv.org/abs/2507.16795>.
- [53] Wenhong Zhu, Ruobing Xie, Rui Wang, Xingwu Sun, Di Wang, and Pengfei Liu. Proximal supervised fine-tuning, 2026. URL <https://arxiv.org/abs/2508.17784>.
- [54] HiLD Workshop Organizers. Workshop on high-dimensional learning dynamics (hild). <https://sites.google.com/view/hidimlearning/home>, 2023. ICML Workshop.
- [55] HiLD Workshop Organizers. Workshop on high-dimensional learning dynamics (hild). <https://sites.google.com/view/hidimlearning/home>, 2024. ICML Workshop.
- [56] HiLD Workshop Organizers. Workshop on high-dimensional learning dynamics (hild). <https://sites.google.com/view/hidimlearning/home>, 2025. ICML Workshop.
- [57] HiLD Workshop Organizers. Workshop on high-dimensional learning dynamics (hild). <https://sites.google.com/view/hidimlearning/home>, 2026. ICML Workshop.

- [58] Sid Black, Gao Leo, Phil Wang, Connor Leahy, and Stella Biderman. Gpt-neo: Large scale autoregressive language modeling with mesh-tensorflow. *Zenodo*, 2021.
- [59] Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, et al. Gpt-neox-20b: An open-source autoregressive language model. In *Proceedings of BigScience Episode# 5–Workshop on Challenges & Perspectives in Creating Large Language Models*, pages 95–136, 2022.
- [60] Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- [61] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- [62] Stella Biderman, Usvsn Prashanth, Lintang Sutawika, Hailey Schoelkopf, Quentin Anthony, Shivanshu Purohit, and Edward Raff. Emergent and predictable memorization in large language models. *Advances in Neural Information Processing Systems*, 36:28072–28090, 2023.
- [63] Curt Tigges, Michael Hanna, Qinan Yu, and Stella Biderman. Llm circuit analyses are consistent across training and scale. *Advances in Neural Information Processing Systems*, 37:40699–40731, 2024.